

2020 second round of Nepal Mathematics Olympiad

October 30, 2022

Problem 1

Prove that there are infinitely many distinct pairs (a, b) of relatively prime positive integers $a > 1$ and $b > 1$ such that $a^b + b^a$ is divisible by $a + b$.

(Source: USAMO 2017)

Problem 2

There are 100 red, 100 yellow and 100 green sticks. One can construct a triangle using any three sticks all of different color. Prove that there is a color such that one can construct a triangle using any three sticks of this color.

(Source: Unknown)

Problem 3

Find all functions $f : R \rightarrow R$ such that

$$f(xf(y) - y^2) = (y + 1)f(x - y)$$

holds for all $x, y \in R$.

(Source: 2019 Baltic Way P3)

Problem 4

Let ABC be a triangle with $\angle ABC = 90^\circ$, and let H be the foot of the altitude from B . The points M and N are the midpoints of the segments AH and CH , respectively. Let P and Q be the second points of intersection of the circumcircle of the triangle ABC with the lines BM and BN , respectively. The segments AQ and CP intersect at the point R . Prove that the line BR passes through the midpoint of the segment MN .

(Source: 2019 Baltic Way P14)

Problem 5

A standard 8×8 chessboard is modified by varying the distances between parallel grid lines, so that the cells are rectangles which are not necessarily squares, and do not necessarily have constant area. The ratio between the area of any white

cell and the area of any black cell is at most 2. Determine the maximum possible ratio of the total area of the white cells to the total area of the black cells.

(Source: Unknown)

Problem 6

Determine all positive integers k such that there exists at least two positive integer solutions (x, y) other than $(1, 1)$ such that

$$\frac{x^3y + x^2y + xy + k}{x^2 + y^2}$$

and

$$\frac{y^3x + y^2x + yx + k}{x^2 + y^2}$$

are both positive integers.

(Source: Unknown)